

Individual Contribution Report

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Project: Performance Evaluation of Vector Indexing Techniques for AI Search Systems

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1. Role & Responsibilities

I serve as the Query & Evaluation Lead for this project. My primary responsibility is to execute the search queries fairly across all indexes and to measure the query-side metrics that define the study — query latency and recall@K against the exact ground truth — and to orchestrate the benchmark so every method is evaluated under identical conditions without caching or ordering bias, aggregating all recorded metrics into the results.

Code Modules Owned

- `query_engine.py` — query execution with per-query timing (mean / p50 / p95) and recall@K against the Flat ground truth
- `benchmark.py` — orchestration that runs every index under identical conditions (single and scalability modes) and aggregates the recorded metrics

Writing Responsibilities

- Research Objectives section of the project proposal
- Evaluation Metrics section of the project proposal
- Risks & Mitigation section of the project proposal
- Evaluation and results-analysis section of the final report

Technical Responsibilities

- Designing the held-out query workload and the unified evaluation pipeline
- Implementing per-query latency timing with warm-up (mean, p50, p95)
- Implementing recall@K measured against the exact Flat-search ground truth
- Orchestrating runs under identical conditions and aggregating all recorded metrics
- Investigating result reproducibility across repeated runs

2. Weekly Contribution Log

The following log records individual actions and outcomes from Week 1 through Week 8, aligned with the course project timeline and phase milestones.

Week 1 — Course Orientation & Background Reading

Actions Taken	- Completed a thorough reading of the course syllabus and grading criteria - Reviewed the Modern Database Architectures module on Brightspace - Conducted independent reading on retrieval evaluation — recall@K, query latency, and benchmarking methodology for ANN systems - Studied how to compare approximate retrieval fairly against an exact baseline
Outcome	Completed Week 1 with a clear grounding in evaluation methodology, ready to lead the query and evaluation lane.

Week 2 — Team Formation, Role Assignment & Proposal Writing

Actions Taken	- Connected with all three team members — Devu, Ashwin, and Nikhil — upon team formation - Participated in the first team meeting to divide responsibilities into four non-overlapping lanes - Took formal ownership of the Query & Evaluation lane (query_engine.py, benchmark.py) - Contributed to setting up the shared repository and GitHub Projects board - Began drafting the Research Objectives and Evaluation Metrics sections of the Pre-Proposal and defined the metric plan
Outcome	Team roles defined with no overlap; evaluation lane owned; the metric plan was drafted.

Week 3 — Pre-Proposal Submission & Evaluation Design

Actions Taken	- Finalized the Research Objectives and Evaluation Metrics sections of the Pre-Proposal - Designed the held-out query workload and the evaluation pipeline skeleton - Defined recall@K measured against the exact Flat-search ground truth - Participated in full-team review and contributed to the on-time Pre-Proposal submission
Outcome	Pre-Proposal submitted; the evaluation methodology and metric definitions were finalized.

Week 4 — Peer Review & Query Engine Implementation

Actions Taken	• Attended the in-person instructor peer review and defended the evaluation metrics and methodology • Incorporated instructor feedback into the Full Proposal draft • Implemented <code>query_engine.py</code> — per-query execution with warm-up and mean / p50 / p95 latency timing • Contributed the Evaluation Metrics section to the Full Proposal
Outcome	The query engine with per-query timing was implemented; the metrics section was drafted.

Week 5 — Full Proposal Submitted & Recall Implementation

Actions Taken	• Finalized the Research Objectives and Evaluation Metrics in the Full Proposal • Implemented <code>recall@K</code> measured against the exact Flat ground truth • Delivered the evaluation-metrics portion in the team's recorded Full-Proposal presentation • Committed the evaluation code to the GitHub repository
Outcome	Full Proposal submitted and presented; <code>recall@K</code> evaluation implemented.

Week 6 — Benchmark Orchestration

Actions Taken	• Implemented <code>benchmark.py</code> — orchestration that runs each index under identical conditions in single and scalability modes • Added aggregation of all recorded metrics into per-method result records • Integrated the evaluation pipeline with the experiment runner • Supported validation of the newly built indexes
Outcome	Benchmark orchestration implemented; per-method metrics aggregated for analysis.

Week 7 — Initial Benchmarking (Flat & IVF)

Actions Taken	• Ran the first benchmarking for the Flat and IVF indexes — collecting latency and recall@K (memory and build-time are recorded by the indexing layer) • Aggregated the per-method metrics for both methods • Verified metric correctness against expected behaviour (Flat = exact, IVF approximate) • Coordinated with the Indexing Lead (Ashwin) on identical test conditions
Outcome	Flat and IVF benchmarked; initial per-method metrics aggregated and verified.

Week 8 — Reproducibility Verification & Metric Aggregation (Current)

Actions Taken	• Extending benchmarking to HNSW as the index becomes available • Investigated and resolved a recall reproducibility issue to ensure consistent results across repeated runs • Aggregated the per-method metrics across all methods for analysis • Began drafting the evaluation and results-analysis section of the final report
Outcome	Recall reproducibility confirmed; per-method metrics aggregated; HNSW benchmarking to follow.

3. Current Project Status

The project is active and ongoing. Phases 1 and 2 are complete and the team is in Phase 3, with HNSW index construction underway and progressing into Phase 4 as initial benchmarking runs for Flat and IVF have begun. My query engine and benchmark orchestration are implemented, and the Flat and IVF metrics have been collected and verified.

Phase 1	Dataset Collection & Preprocessing	Complete	ABC News Headlines ingested, cleaned, normalized, and segmented using <code>data_ingestion.py</code> and <code>prepare_headlines.py</code> .
Phase 2	Embedding Generation	Complete	Deterministic 384-dimensional Sentence-BERT embeddings generated via <code>embeddings.py</code> for all corpus sizes.
Phase 3	Index Construction	In Progress	Flat and IVF indexes built and tested; HNSW index currently under construction in FAISS.
Phase 4	Benchmarking & Evaluation	In Progress	Initial benchmarking runs begun for Flat and IVF; HNSW benchmarking to follow as index construction completes.
Phase 5	Analysis & Documentation	Not Started	Report writing, results analysis, and presentation preparation scheduled over the remaining weeks.

Phase 4 — Benchmarking & Evaluation (My Core Phase)

My query engine executes the held-out queries against each index with warm-up and per-query timing (mean, p50, p95) and computes recall@K against the exact Flat ground truth. `benchmark.py` orchestrates these runs under identical conditions and aggregates the recorded metrics (the query-side latency and recall, together with the memory and build-time reported by the indexing layer). Initial Flat and IVF metrics are collected and verified; HNSW benchmarking will follow as its construction completes.

Remaining Work

My remaining tasks are to complete HNSW benchmarking, finalize the aggregated per-method metrics, and write the evaluation and results-analysis section of the final report.